

Security aspects of webcams and CCTVs

Nowadays, there are umpteen smart gadgets and webcams which are connected on IoT environments with IP addresses. These IP based webcams are indexed by specialised search engines. Shodan (shodan.io) is one of the famous search engines that indexes and saves the information of IP based webcams. By simply installing the webcams, the security and integrity cannot be ensured unless the security of the IP addresses is ensured. Shodan is able to fetch information about webcams and IoT devices that are vulnerable.

On creating an account on Shodan, an API key is generated from the URL <https://account.shodan.io>, and it can be used for fingerprinting and deep information about IoT based devices.

The official portal of Shodan provides the API key that can be integrated with Python code and the fingerprinting of a webcam or IoT device can be done; however, this is quite unsafe if the IoT device is not secured with a strong password.

```
import shodan
SHODAN_API_KEY =
"*****"

shodanapi = shodan.Shodan(SHODAN_API_KEY)
thishost = shodanapi.host('Target IP Address')
print("""
    Extracted IP: {}
    Extracted Organization: {}
    Extracted Operating System: {}
    """).format(thishost['ip_str'], host.get('org', 'n/a'), thishost.get('os', 'n/a'))
for item in thishost['mydata']:
    print("""
        Extracted Port: {}
        Extracted Banner: {}
        """).format(item['port'], item['mydata']))
```

If the script is executed, port scanning and deep information of the

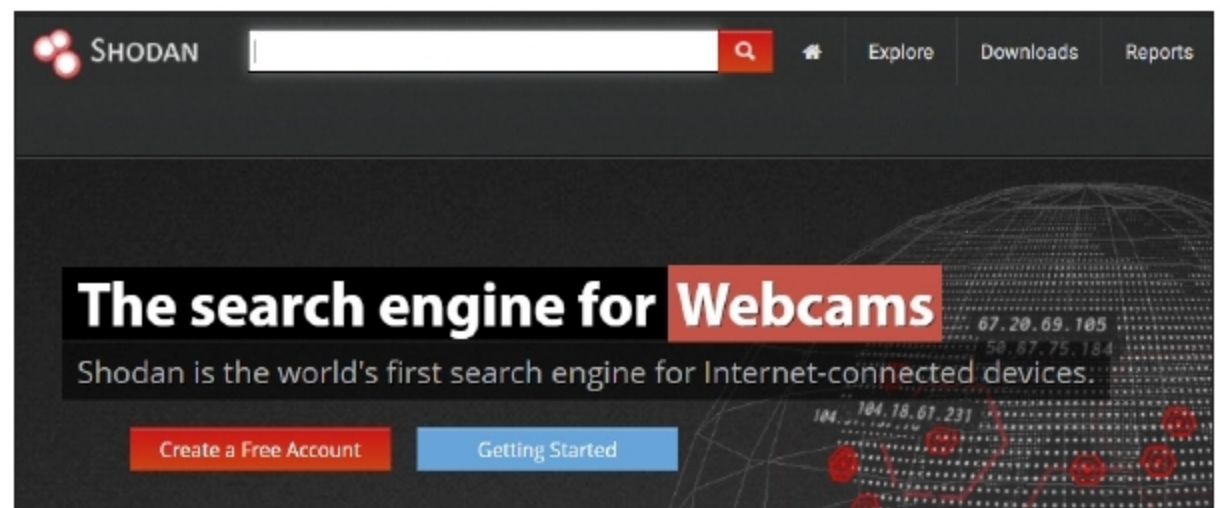


Figure 3: Shodan search engines for webcams and IoT devices

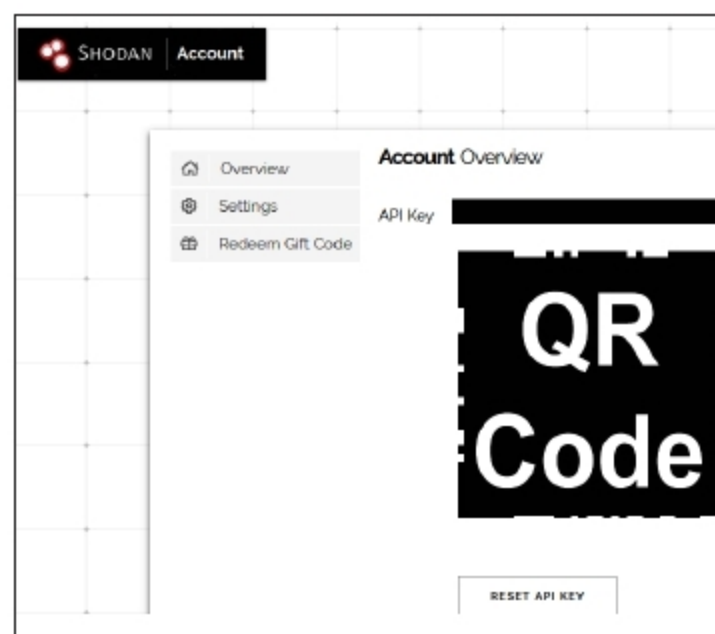


Figure 4: API key and QR code by Shodan

IP webcam can be extracted using a Python Script which is available as free and open source. Therefore, the immense vulnerability of webcams and

IoT gadgets must be secured.

To ensure the security and privacy of the webcams and IoT gadgets, strong passwords should be used, otherwise anybody can access them.

With the adoption and integration of advanced devices on the Web environment, there is need to ensure the security and integrity to Internet connected infrastructure. Apart from Python programming, blockchain technology, quantum

cryptography and multi-factor authentication are the other secured approaches that are very important for real world applications. **END** 🐧

References

- [1] NEWS18: '1,852 Cyber Attacks Hit India Each Minute Last Year; Mumbai, Delhi Most Affected', URL: <https://www.news18.com/news/tech/1852-cyber-attacks-hit-india-each-minute-last-year-mumbai-delhi-most-affected-2295963.html> September 3, 2019
- [2] INC42: 'India Second Most Affected Country Due To Cyber Attacks: Report', URL: <https://inc42.com/buzz/cyber-attacks-india/>, 03 May'19
- [3] STATISTA: 'Cyber crime in India - Statistics & Facts', URL: <https://www.statista.com/topics/5054/cyber-crime-in-india/>
- [4] Economic Times: 'Cyber attacks grew 22 per cent on India's IoT deployments in Q2', URL: <https://economictimes.indiatimes.com/tech/internet/cyberattacks-grew-22-on-indias-iot-deployments-in-q2/articleshow/70606639.cms?from=mdr>, Aug 09, 2019

By: Dr Gaurav Kumar

The author is the founder director of Magma Research and Consultancy Services. He is associated with various universities, institutes and autonomous organisations for the delivery of expert lectures, technical workshops and consultancy on the latest technologies and tools.